7 **(a)** **(i)** Describe the *photoelectric effect* in terms of energy.

When electromagnetic radiation is incident upon the surface of a material, if its frequency exceeds the threshold frequency of the material, then it is able to eject electrons from the surface. [1] Part of the energy of the incident photon is used to remove an electron and the remaining energy appears as the kinetic energy of the ejected electron. [1]

....[2]

(ii) Explain one way in which the photoelectric effect provides evidence for the particulate nature, and not wave nature, of electromagnetic radiation.

Experimental observations show that no electrons are emitted unless the frequency of the monochromatic incident light is greater than a minimum (threshold) value regardless of the light intensity. On the contrary, wave theory predicts that the photoelectric effect should occur for any frequency of the incident light, and that its intensity can be increased to cause photoelectron emission If the light frequency is too low. [1] This observation suggests that the energy of the incident light has a particulate nature in that each photon has a fixed amount of energy. If frequency of the incident photon is less than the threshold frequency of the target metal, energy of the incident photon is insufficient to overcome the work function of the metal. Increasing the intensity would only increase the number of photons arriving per unit time, but the energy of each photon remains insufficient to cause photoelectric effect. [1]

(b) The graph drawn in Fig. 7.1 shows how the maximum kinetic energy E_k of a photoelectron from a particular material varies with the frequency f of the electromagnetic radiation that causes the emission of photoelectrons.

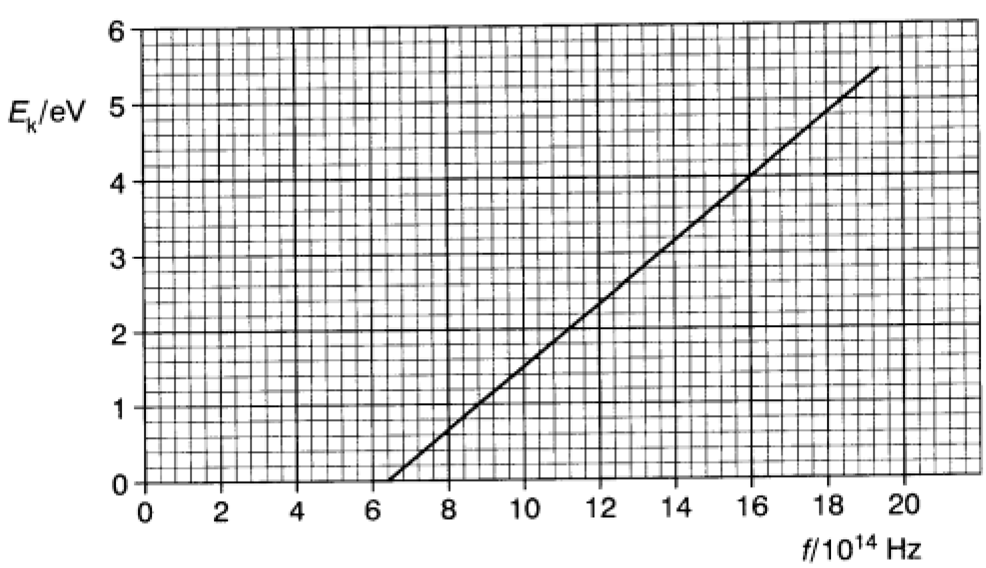

Fig. 7.1

(i) Use the graph to determine

1. the threshold frequency for this material,

At the threshold frequency, electrons are emitted with near-zero KE.
So, threshold frequency = $\underline{6.4 \times 10^{14}\text{ Hz}}$

threshold frequency = ... Hz [1]

2. the maximum kinetic energy of photoelectrons from this material when it is illuminated with electromagnetic radiation of frequency 18.0×10^{14} Hz.

From graph, when f = 18.0×10^{14} Hz,
Max KE = 4.8 eV [1]
= $4.8 \times 1.6 \times 10^{-19}$ J
= 7.68×10^{-19} J [1]

OR
Using the photoelectric equation,
Photon energy = Work function + max. KE of electrons
$hf = \phi + KE_{max}$
At threshold frequency,
work function = photon energy
$\phi = hf_0$
$hf = hf_0 + KE_{max}$
$6.63 \times 10^{-34} \times 18.0 \times 10^{14} = 6.63 \times 10^{-34} \times 6.4 \times 10^{14} + KE_{max}$
KE_{max} = $\underline{7.7 \times 10^{-19}\ \text{J}}$

maximum kinetic energy = ... J [2]

(ii) Determine the minimum potential difference between the electrodes in the photoelectric experiment that is needed to reduce the photocurrent to zero.

Loss in KE = Gain in electric PE
$\Delta KE = q\Delta V$
$7.7 \times 10^{-19} = 1.6 \times 10^{-19} \times \Delta V$
ΔV = $\underline{4.8\ \text{V}}$

minimum potential difference = ... V [2]

(c) Electromagnetic waves have a wave nature as well as a particulate nature. This is known as the wave-particle duality. Describe an experiment in which particles exhibit wave nature.

Possible answers:
- When a beam of electrons passes through a thin film of graphite, concentric rings are seen on a fluorescent screen, revealing the occurrence of interference of electron matter waves.
- When a beam of electrons is reflected from a surface of atoms arranged in a regular pattern, constructive interference is observed in certain directions, showing that the electrons are behaving as waves as they interact with the atoms.

[1]

[Total: 10]

Section B

Answer **one** question from this section in the spaces provided.

8 (a) State what is meant by the *binding energy* of a nucleus and how it is related to the mass